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(54) **BATTERY PACK INCLUDING CAP CONFIGURED TO SUPPORT CARTRIDGE ASSEMBLIES HOLDING AGENTS**

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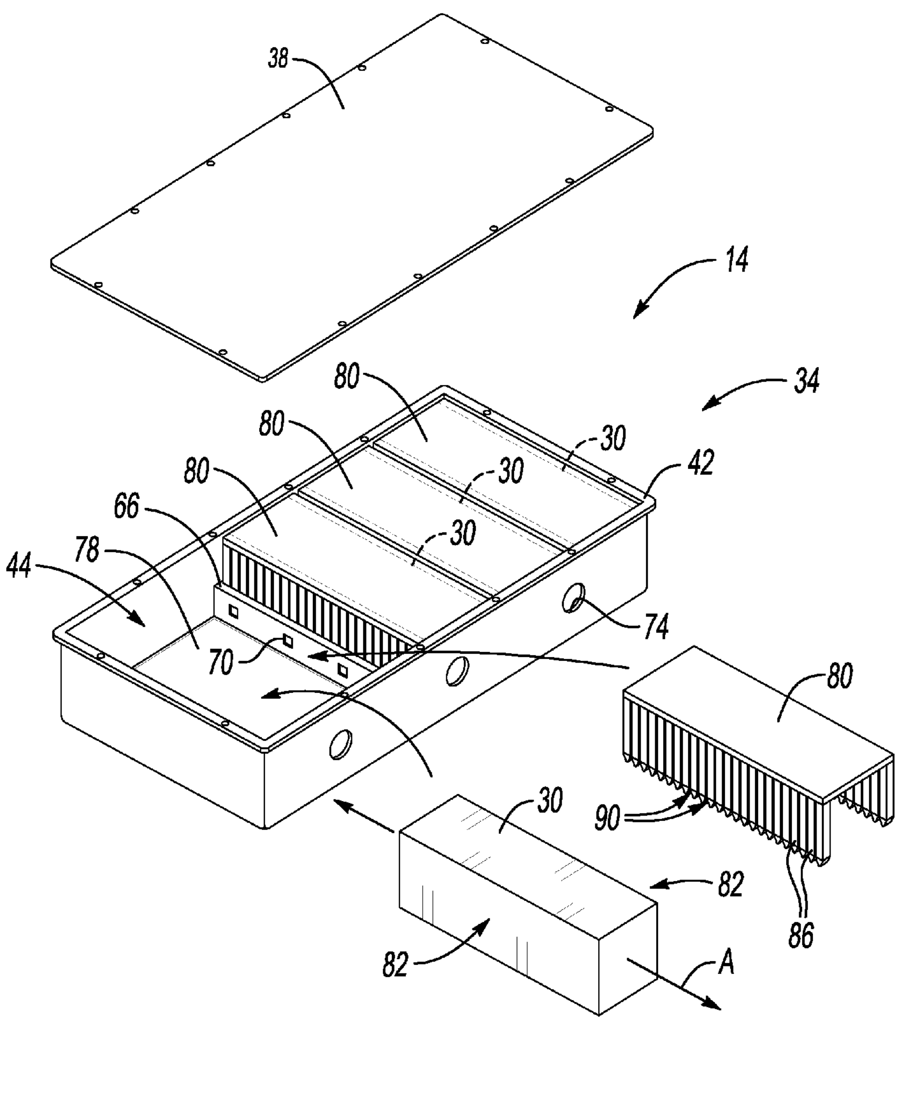
(57) **ABSTRACT**

The techniques described herein relate to a traction battery pack assembly, including an enclosure assembly that provides an interior area and a cell stack within the interior area. The cell stack includes a plurality of battery cells disposed along a cell stack axis. Each battery cell includes at least one terminal tab that projects outward from the cell stack axis. Further included is a cap and a plurality of fingers projecting from the cap. Each of the fingers is spaced-apart from one another to provide at least one slot that receives a portion of the at least one terminal tab. Each of the plurality of fingers is provided by a cartridge assembly. Each of the cartridge assemblies holds agents and is configured to release the agents in response to a thermal event proximate the respective cartridge assembly.



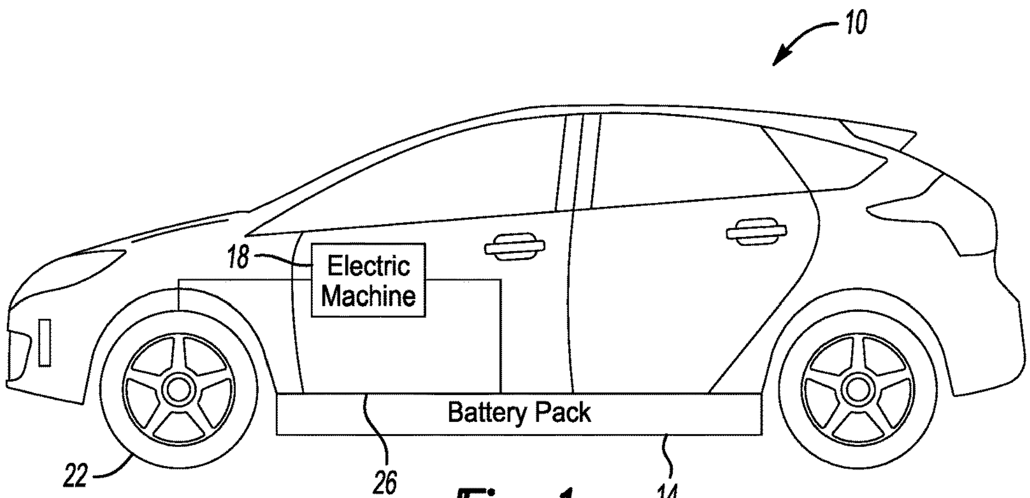
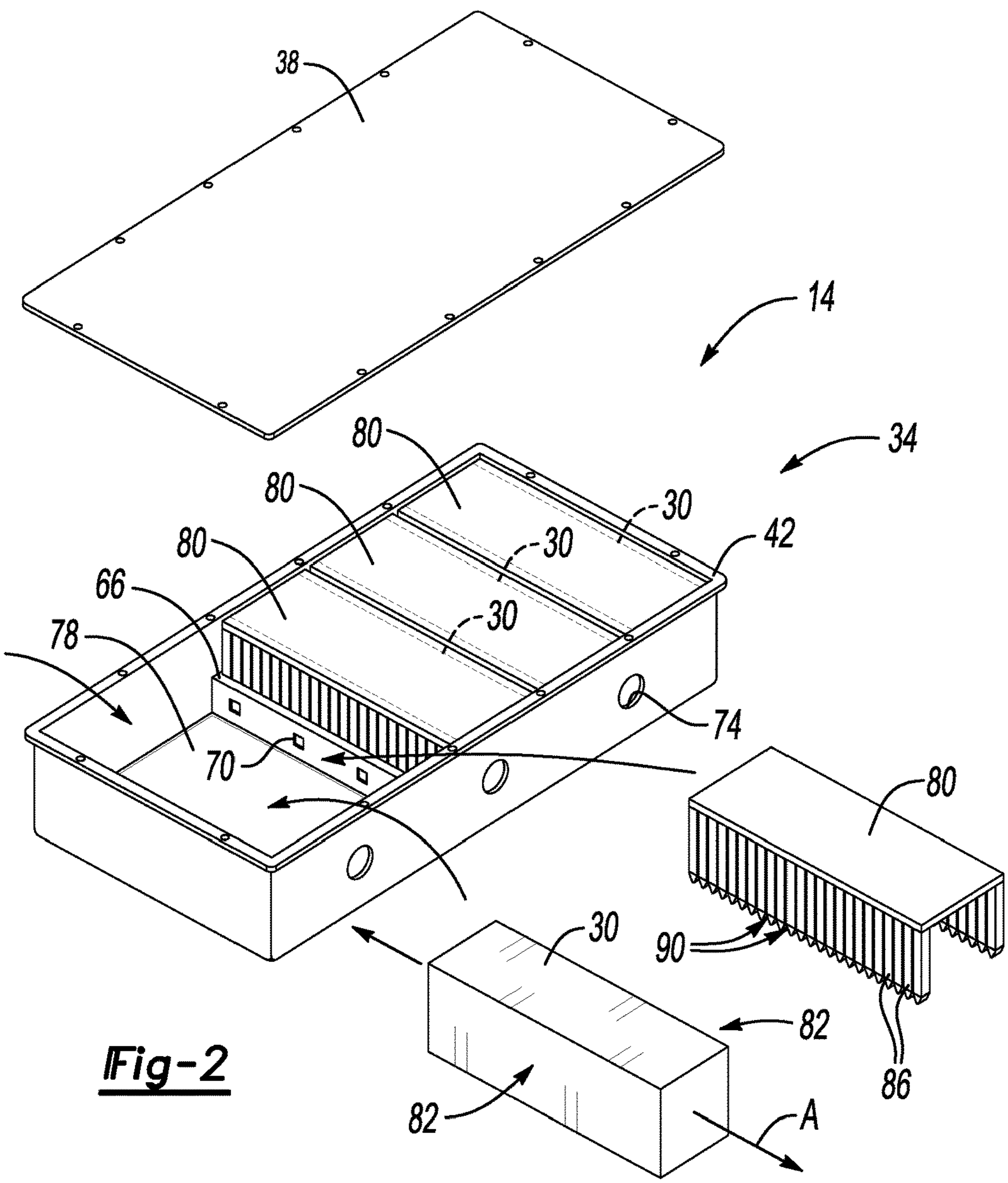

10
18
Electric
Machine
Battery Pack
22
26
14

Fig-1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14
34
80
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80
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A

Fig-2

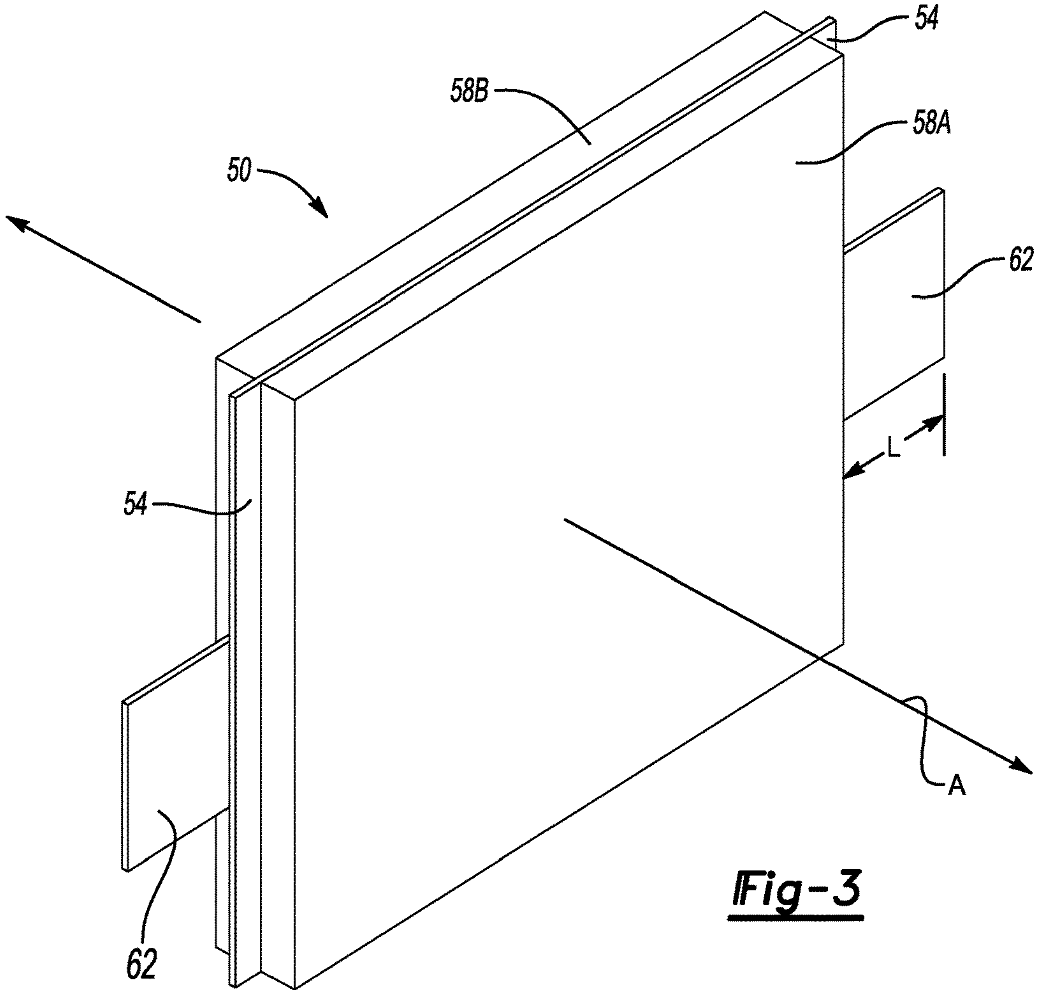

Fig-3

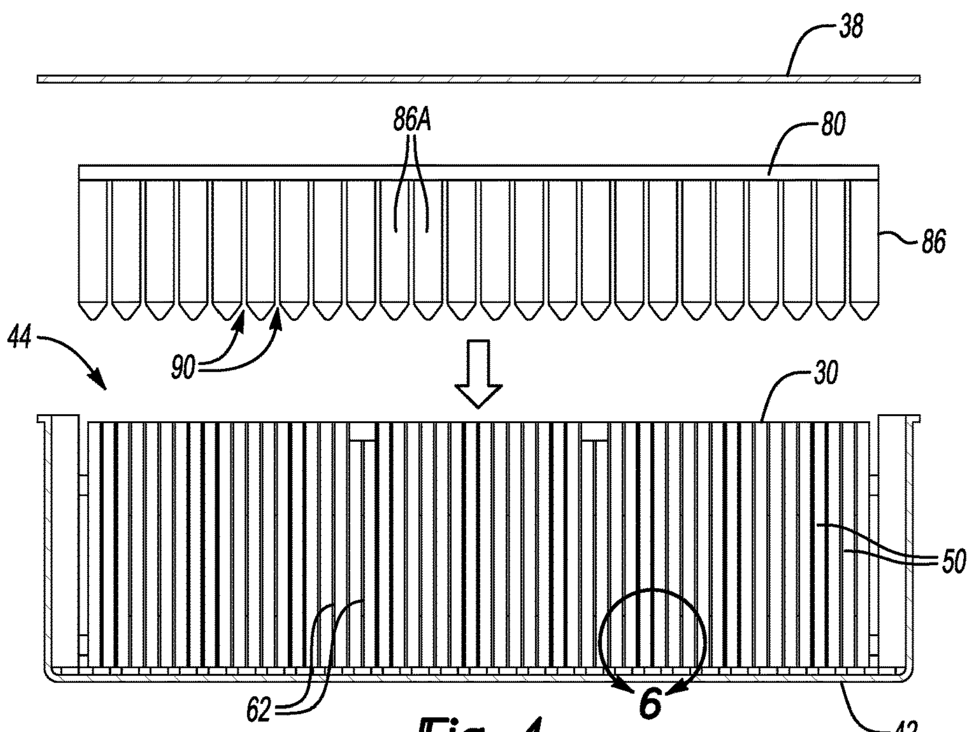

Fig-4

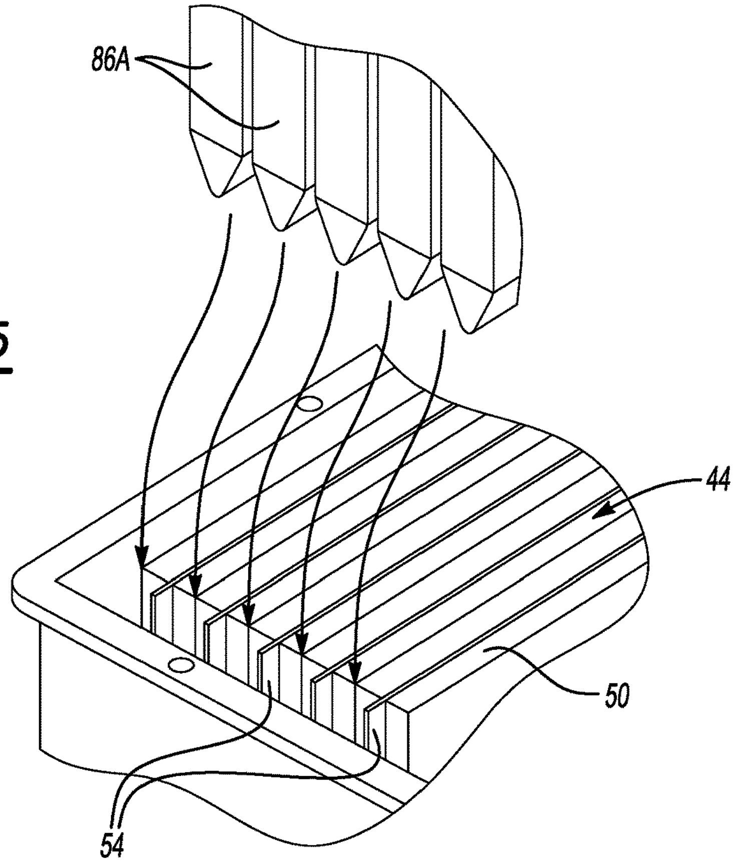

Fig-5

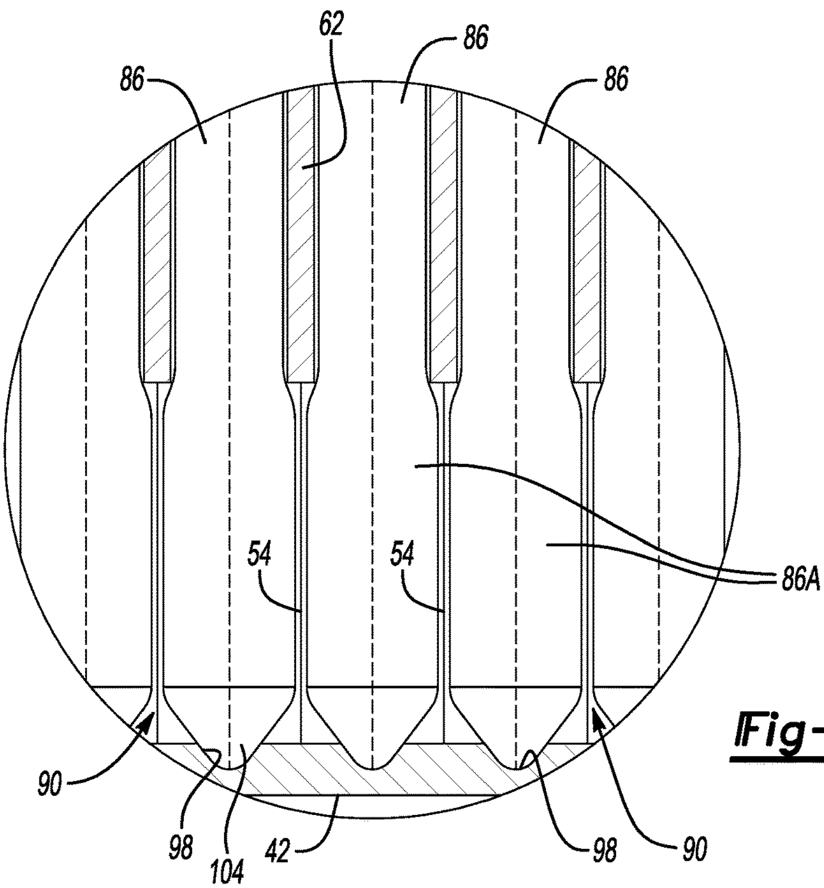

Fig-6

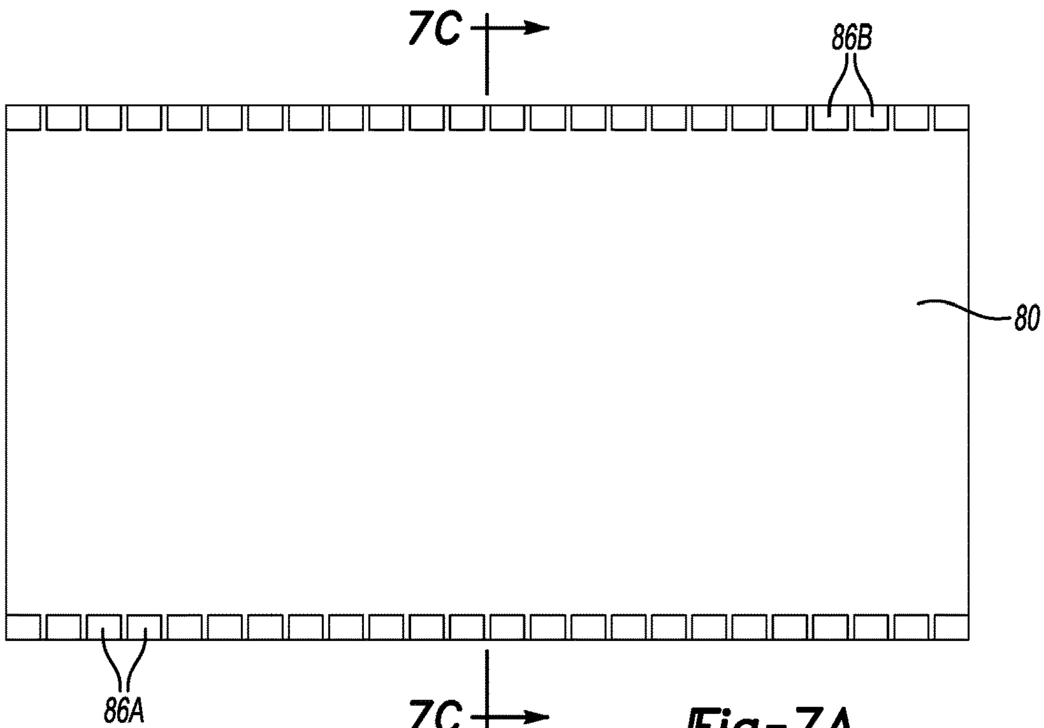

Fig-7A

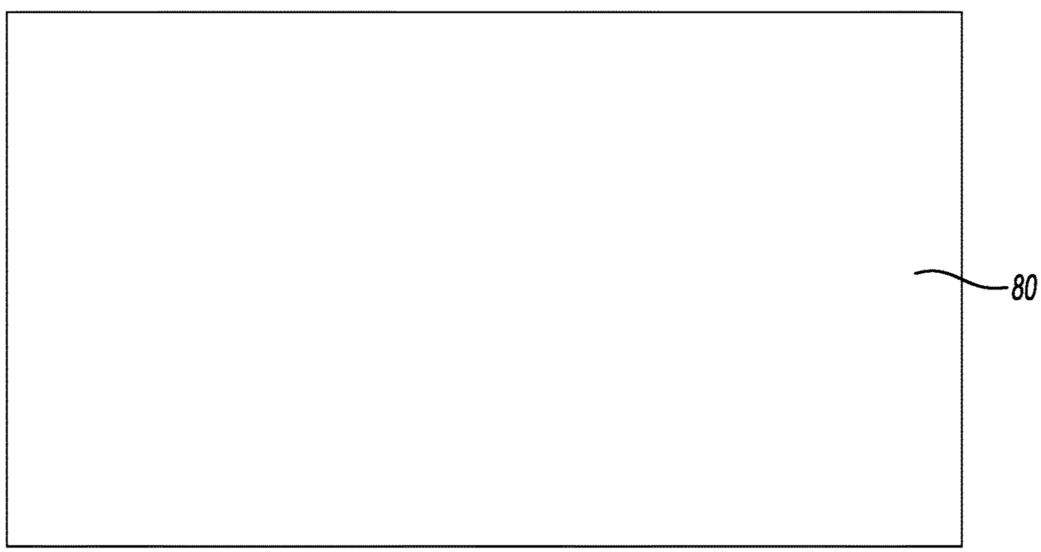

Fig-7B

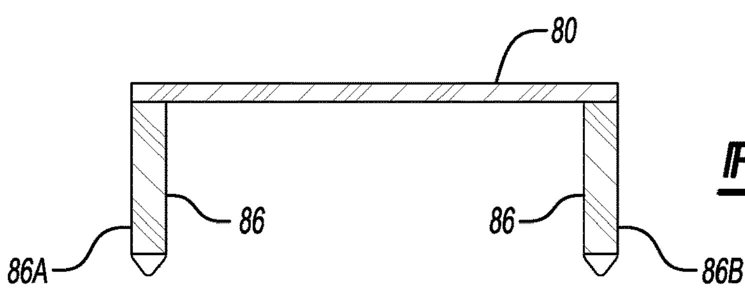

Fig-7C

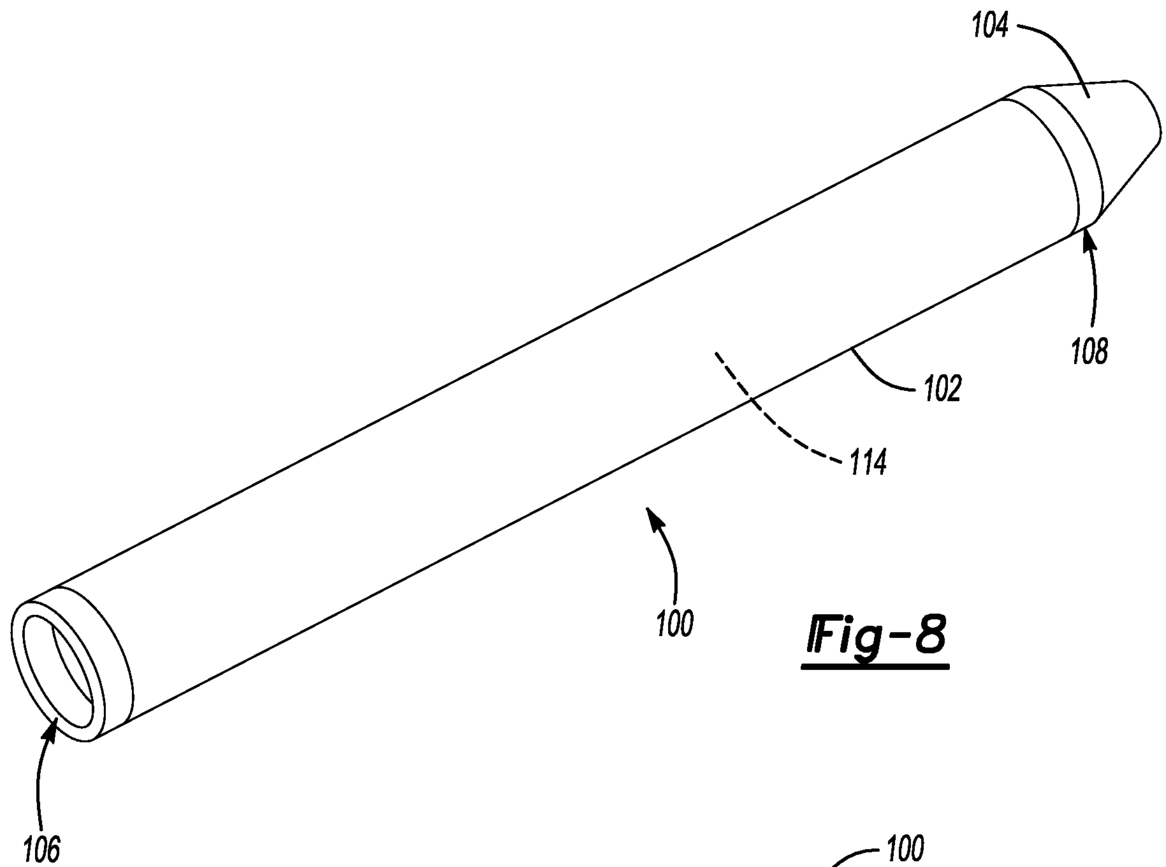

Fig-8

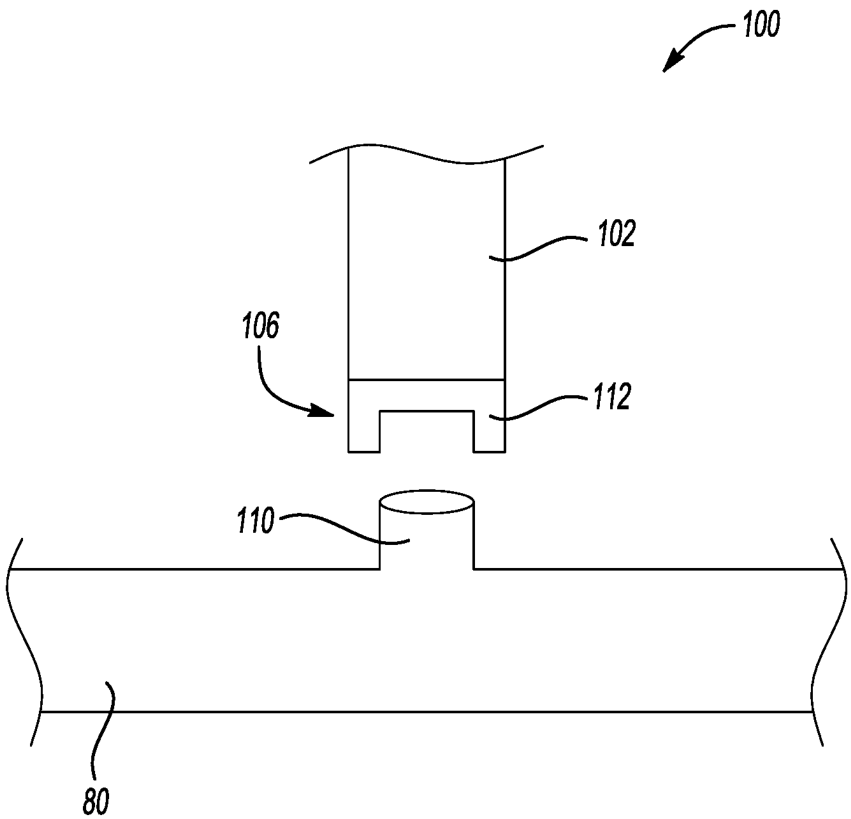

Fig-9

BATTERY PACK INCLUDING CAP CONFIGURED TO SUPPORT CARTRIDGE ASSEMBLIES HOLDING AGENTS

TECHNICAL FIELD

[0001] This disclosure relates generally to a traction battery pack for an electrified vehicle. In particular, this disclosure relates to a cap configured to support a plurality of cartridge assemblies relative to the traction battery pack. The cartridge assemblies hold agents, and are configured to release those agents during a thermal event. A corresponding method is also disclosed.

BACKGROUND

[0002] A high voltage traction battery pack typically powers the electric machines and other electrical loads of an electrified vehicle. The traction battery pack includes a plurality of battery cells. The traction battery pack can, from time to time, experience a thermal event where one or more of the battery cells vent and expel battery vent byproducts. The battery thermal event may occur due to, for example, over-charging conditions, over-discharging conditions, or other conditions. The vent byproducts can include gases and effluent particles. Thermal energy within the vent byproducts can cause the thermal event to cascade to other battery cells.

SUMMARY

[0003] In some aspects, the techniques described herein relate to a traction battery pack assembly, including: an enclosure assembly that provides an interior area; a cell stack within the interior area, wherein the cell stack includes a plurality of battery cells disposed along a cell stack axis, wherein each battery cell within the plurality of battery cells includes at least one terminal tab that projects outward from the cell stack axis; a cap; and a plurality of fingers projecting from the cap, wherein each of the fingers is spaced-apart from one another to provide at least one slot that receives a portion of the at least one terminal tab, wherein each of the plurality of fingers is provided by a cartridge assembly, wherein each of the cartridge assemblies holds agents, and wherein each of the cartridge assemblies is configured to release the agents in response to a thermal event proximate the respective cartridge assembly.

[0004] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the agents are configured, when released from a respective cartridge assembly, to do one or more of (i) arrest oxygen, (ii) electrically isolate, and (iii) reduce temperature.

[0005] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein each of the cartridge assemblies holds a mixture of silicon dioxide, aluminum oxide, and sodium silicate.

[0006] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the cap is an integrally-formed structure without any joints or seams.

[0007] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein each of the plurality of fingers is formed separately from one another and formed separately from the cap.

[0008] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the plurality of fingers includes at least one first finger on a first side of the cell stack, and at least one second finger on an opposite, second side of the cell stack.

[0009] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the cap is disposed alongside a third side of the cell stack.

[0010] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the third side is transverse to both the first side and the second side.

[0011] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein each of the cartridge assemblies includes: a hollow tube, and a container cap adjacent an end of the hollow tube opposite the cap.

[0012] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein each of the cartridge assemblies is configured such that melting of either the hollow tube or the container cap releases the agents within the respective cartridge assembly.

[0013] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein: the cap includes a plurality of male components, and each of the hollow tubes includes a female component press-fit relative to a corresponding one of the male components.

[0014] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the plurality of fingers are a plurality of first fingers disposed along a first side of the cell stack and a plurality of second fingers disposed along a second side of the cell stack.

[0015] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the cell stack is a first cell stack, and further including at least one second cell stack within the interior area.

[0016] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein the plurality of battery cells are a plurality of pouch-style battery cells, each pouch-style battery cell having a crimped edge that projects outward from the cell stack axis about a circumferential perimeter of the pouch-style battery cell, wherein the at least one slot receives a portion of at least one of the crimped edge.

[0017] In some aspects, the techniques described herein relate to a traction battery pack assembly, wherein each of the terminal tabs within the plurality of terminal tabs extends through one of the crimped edges.

[0018] In some aspects, the techniques described herein relate to a method, including: placing a cell stack within an interior area of an enclosure of a battery pack; inserting an assembly including a cap and a plurality of fingers into the interior area, wherein each of the fingers is provided by a cartridge assembly, wherein each of the cartridge assemblies holds agents, and wherein each of the cartridge assemblies is configured to release the agents in response to a thermal event proximate the respective cartridge assembly; and during the inserting step, receiving at least one tab terminal of the cell stack within a slot between a first finger of the plurality of fingers and a second finger of the plurality of fingers.

[0019] In some aspects, the techniques described herein relate to a method, wherein the agents are configured, when released from a respective cartridge assembly, to do one or more of (i) arrest oxygen, (ii) electrically isolate, and (iii) reduce temperature.

[0020] In some aspects, the techniques described herein relate to a method, wherein each of the cartridge assemblies holds a mixture of silicon dioxide, aluminum oxide, and sodium silicate.

[0021] In some aspects, the techniques described herein relate to a method, wherein the cap is an integrally-formed structure without any joints or seams.

[0022] In some aspects, the techniques described herein relate to a method, wherein the plurality of fingers includes a plurality of fingers on a first side of the cell stack and a plurality of fingers on an opposite, second side of the cell stack.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. **1** illustrates a side view of an example electrified vehicle.

[0024] FIG. **2** illustrates an expanded, perspective view of a battery pack from the electrified vehicle of FIG. **1** according to an exemplary aspect of the present disclosure.

[0025] FIG. **3** illustrates a battery cell from the battery pack of FIG. **2**.

[0026] FIG. **4** illustrates a section view of the battery pack of FIG. **2** as an arrangement including a cap and a plurality of fingers is installed.

[0027] FIG. **5** illustrates a perspective view of a corner portion of the battery pack of FIG. **2** as the arrangement including the cap and the plurality of fingers is installed.

[0028] FIG. **6** illustrates a close-up view of an area of FIG. **4**.

[0029] FIGS. **7**A to **7**C illustrate various views of the arrangement of the cap and the plurality of fingers.

[0030] FIG. **8** illustrates an example cartridge assembly.

[0031] FIG. **9** illustrates an example interface between a cartridge assembly and the cap.

DETAILED DESCRIPTION

[0032] This disclosure relates generally to a traction battery pack for an electrified vehicle. In particular, this disclosure relates to a cap configured to support a plurality of cartridge assemblies relative to the traction battery pack. The cartridge assemblies hold agents, and are configured to release those agents during a thermal event. A corresponding method is also disclosed. The arrangement of the cap relative to the cartridge assemblies facilitates assembly and disassembly of the cartridge assemblies relative to the traction battery pack. These and other features are discussed in greater detail in the following paragraphs of this detailed description.

[0033] With reference to FIG. **1**, an electrified vehicle **10** includes a battery pack **14**, an electric machine **18**, and wheels **22**. The battery pack **14** powers an electric machine **18**, which can convert electrical power to mechanical power to drive the wheels **22**.

[0034] The battery pack **14** is, in the exemplary embodiment, secured to an underbody **26** of the electrified vehicle **10**. The battery pack **14** could be located elsewhere on the electrified vehicle **10** in other examples.

[0035] The electrified vehicle **10** is an all-electric vehicle. In other examples, the electrified vehicle **10** is a hybrid electric vehicle, which selectively drives wheels using torque provided by an internal combustion engine instead of, or in addition to, an electric machine. Generally, the electrified vehicle **10** could be any type of vehicle having a traction battery pack.

[0036] With reference now to FIGS. **2** and **3**, the battery pack **14** includes a plurality of cell stacks **30** held within an enclosure assembly **34**. In the exemplary embodiment, the enclosure assembly **34** includes an enclosure cover **38** and an enclosure tray **42**. The enclosure cover **38** is secured to the enclosure tray **42** to provide an interior area **44** that houses the cell stacks **30**. The enclosure cover **38** can be secured to the enclosure tray **42** using mechanical fasteners (not shown), for example.

[0037] Each of the cell stacks **30** includes a plurality of battery cells **50** (or simply, "cells") disposed along a respective cell stack axis A. Within each cell stack **30**, the battery cells **50** are stacked side-by-side relative to each other along the cell stack axis A. The cells **50** are shown in highly schematic form in the cell stacks **30** of FIG. **2**.

[0038] The cells **50** can store and supply electrical power. Although specific numbers of the cell stacks **30** and cells **50** are illustrated in the various figures of this disclosure, the battery pack **14** could include any number of the cell stacks **30** having any number of individual battery cells **50**.

[0039] In this exemplary embodiment, the battery cells **50** are lithium-ion pouch cells. However, battery cells having other geometries (cylindrical, prismatic, etc.) other chemistries (nickel-metal hydride, lead-acid, etc.), or both could alternatively be utilized within the scope of this disclosure.

[0040] The example battery cells **50** include a crimped edge **54** where a first case **58**A of the battery cell **50** is joined to a second case **58**B of the battery cell **50**. The crimped edge **54** can extend partially or completely about a circumferential perimeter of the associated battery cell **50**. Terminal tabs **62** of the battery cells **50** project outward away from the cell stack axis A through the crimped edge **54**. When the battery cell **50** is within the cell stack **30**, the terminal tabs **62** extend outward from the cell stack axis A and can connect to a busbar, for example.

[0041] From time to time, pressure and thermal energy within one or more of the battery cells **50** can increase. The pressure and thermal energy increase can be due to an overcharge condition, for example. The pressure and thermal energy increase can cause the associated battery cell **50** to rupture and expel vent byproducts, such as gas and debris, from within the battery cell **50**.

[0042] The vent byproducts can be released from the associated battery cell **50** through a ruptured area of the associated battery cell **50**. The vent byproducts may be released through a ruptured area of the crimped edge **54**. The vent byproducts could also be released through a designated vent within the one or both of the first case **58**A and the second case **58**B. The designated vent could be a membrane that yields in response to increased pressure.

[0043] The battery pack **14**, in these examples, includes cross-member assemblies **66** disposed between cell stacks **30**. The example cross-member assemblies **66** extend longitudinally in a direction that is parallel to the cell stack axes A. The cross-member assemblies **66** and the cell stack axes A extend in a cross-vehicle direction (i.e., from a driver side to a passenger side). Bus bars that connect to the terminal tabs **62** can be mounted to the cross-member assemblies **66**.

[0044] In this example, the cross-member assemblies **66** could include venting passageways. Vent byproducts from one or more of the battery cells **50** can move through at least one of the openings **70** into the venting passageway. The vent byproducts are communicated though the venting passageway through an enclosure vent **74** to an area outside the battery pack **14**. The openings **70**, the enclosure vent **74**, or both can be covered by respective membranes, for example, when not venting. During venting, the vent byproducts can rupture the membranes so that the vent byproducts can flow from the battery cells **50**, through the openings **70** to the venting passageway and then through the enclosure vent **74**.

[0045] In another example, vent byproducts from one or more of the battery cells **50** is routed through one or move vents in the enclosure cover **38** of the traction battery pack **14**. Other methods of conveying vent byproducts from the traction battery pack **14** are possible and come within the scope of this disclosure.

[0046] The battery pack **14** includes a plurality of thermal barriers **78** within the interior area **44**. The thermal barriers **78** include portions disposed along a bottom of the cell stacks **30**, as generally shown in FIG. **2**. The thermal barriers **78** block thermal energy from swirling back against other battery cells **50** that are not venting. A thermal barrier **78** is shown relative to a bottom of the interior area **44** in FIG. **2**, the thermal barriers **78** could alternatively or additionally be provided adjacent a bottom or a side of the interior area **44**, in other examples.

[0047] With reference now to FIGS. **4** to **7**C, and continuing reference to FIGS. **2** and **3**, this disclosure includes an arrangement including a cap **80** and a plurality of fingers **86** spaced-apart from each other to provide a plurality of slots **90**. The cap **80** is an integrally-formed structure without any joints or seams, in one example. The cap **80** is made of a metallic material, in one example. The cap **80** is configured to interface with each of the fingers **86**. In an example, each of the fingers **86** is configured to be press-fit relative to the cap **80**. The fingers **86** are formed separately from the cap **80** and separately from one another.

[0048] While the cap **80** is shown separately from the thermal barrier **78**, the cap **80** could be provided by the thermal barrier **78**, in another embodiment. The thermal barrier **78** may be a plastic insulator of the battery pack **14**.

[0049] In an embodiment, the fingers **86** are configured to be attached to the cap **80**. In that state, the cap **80** and fingers **86** are moveable as an assembly. The cap **80** and fingers **86** can be installed relative to the battery pack **14** as an assembly. When the cap **80** and fingers **86** are installed within the battery pack **14**, the slots **90** each receive a portion of one or more of the terminal tabs **62** that extend from one or more of the battery cells **50**. The slots **90** additionally receive a portion of the crimped edge **54** from one or more of the battery cells **50**.

[0050] In this example, a plurality of first fingers **86**A (FIG. **7**A) project from a first side of the cap **80** and a plurality of second fingers **86**B project from a second side of the cap **80** opposite the first side. The first fingers **86**A can be placed along one of the outboard sides **82** (FIG. **2**) of one of the cell stacks **30**, and the plurality of second fingers **86**B can be placed along an opposing outboard side of that cell stack **30**. The plurality of first fingers **86**A and the plurality of second fingers **86**B each project from the cap **80**, which is disposed alongside another side of the cell stack **30**—here an upper side of the cell stack **30**. The outboard sides **82** are substantially parallel to each other in this example. The third side is transverse, here perpendicular, to the outboard sides **82**. A section of the cap **80** taken through the one of the first fingers **86**A and one of the second fingers **86**B has a substantial "C" shape as shown in FIG. **7**C.

[0051] Each of the first fingers **86**A and second fingers **86**B are provided by individual cartridge assemblies. An example cartridge assembly **100** is shown in FIG. **8**. The cartridge assembly **100** includes a hollow tube **102** and a cartridge cap **104**. The hollow tube **102** includes a first end portion **106** configured to interface with the cap **80**, and a second end portion **108** attached to the cartridge cap **104**.

[0052] An example interface between the cap **80** and the cartridge assembly **100** is shown in FIG. **9**. As shown in FIG. **9**, the cap **80** includes a male component **110** configured to interface with a female component **112**, which is mounted adjacent the first end portion **106**. The male and female components **110**, **112** facilitate a press-fit between the cartridge assembly **100** and the cap **80**.

[0053] In this example, the cartridge cap **104** is tapered leading to its end, which helps to guide the first fingers **86**A and second fingers **86**B into an installed position.

[0054] The enclosure tray **42** can include recessed features **98** (FIG. **6**) that each receive part of one of the tapered end portions of the cartridge caps **104**. Receiving the tapered end portion of the cartridge cap **104** within the recessed features **98** can help to align and locate the fingers **86** during assembly.

[0055] In this example, the hollow tube **102** of each cartridge assembly **100** provides an interior volume. The cartridge assembly **100** holds a agents, namely a mixture of agents **114**, within the volume. The mixture of agents can suppress thermal cascades.

[0056] The hollow tube **102** and the cartridge cap **104** can be made of materials such as silicone, thermoplastic vulcanisate (TPV), polyethylene, acrylic, or some combination of these. The hollow tube **102**, the cartridge cap **104**, or both, are designed to melt when exposed to temperatures that, in this example, range from 150 to 250 Celsius. Such temperatures can be present during a thermal event proximate a respective cartridge assembly **100**, including when one or more of the battery cells adjacent to the cartridge assembly **100** is venting vent byproducts. Melting the hollow tube **102**, the cartridge cap **104**, or both, releases the mixture of agents held within the interior volume of the cartridge assembly **100**.

[0057] The mixture of agents that are released can include endothermic materials and materials that help to electrically isolate. Example materials can include sodium silicate, a ceramic compound, melamine poly (zinc phosphate), aluminum tri-hydrate, and silicon dioxide. Other potential agents could include silica, mica, basalt, aerogels, etc.

[0058] The sodium silicate can help to absorb thermal energy. The sodium silicate can be in granular form. The granules of sodium silicate can have a diameter that is from 5 to 100 microns. In an example embodiment, from 25 to 40% of the mixture of agents is sodium silicate.

[0059] The ceramic-based compound can include zirconium dioxide, aluminum oxide, and silicon dioxide. The ceramic-based compound can have the form of beads that have a diameter ranging from 2 to 3 millimeters. In the example embodiment, 25 to 40% of the mixture of agents is ceramic-based beads. The ceramic compound can include more than 10% zirconium dioxide, more than 45% with aluminum oxide and more than 40% of silicon dioxide. The aluminum oxide can help to electrically isolate during the thermal event.

[0060] In other examples, the ceramic-based compound is 100% silicon dioxide and the zirconium dioxide and aluminum oxide are omitted.

[0061] The mixture of agents can include additional aluminum oxide (outside the ceramic-based compound) that acts as a filler agent between the beads of the ceramic based compound. The additional aluminum oxide can be particles having diameters ranging from 5 to 30 microns. The additional aluminum oxide can “fill in the gaps” between larger beads and particles within the mixture of agents. Again, the aluminum oxide can help to electrically isolate during the thermal event. In the example embodiment, 5 to 10% of the mixture of agents is filler aluminum oxide.

[0062] The melamine poly (zinc phosphate) can help to generate nitrogen gas to arrest oxygen generated from electrodes during the thermal event. Reducing available oxygen can help to suppress the thermal event. In the example embodiment, 5 to 8% of the mixture of agents is melamine poly (zinc phosphate). The melamine poly (zinc phosphate) can catalyze formation of a layer that supports an effective intumescent effect. The melamine poly (zinc phosphate) can act as a synergist to support suppression of airborne particulates during the thermal event. The melamine poly (zinc phosphate) can act as a heat sink due to endothermal decomposition. The melamine poly (zinc phosphate) can provide electrical isolation during the thermal event particularly with respect to the vent byproducts and surrounding components.

[0063] The aluminum tri-hydrate can endothermically react during the thermal event to help to reduce temperatures during the thermal event, particularly temperatures of the vent byproducts. In an example embodiment, 3 to 5% of the mixture of agents is aluminum tri-hydrate.

[0064] In other examples, the aluminum tri-hydrate is omitted from the mixture of agents and is not included within the cartridge assembly **100**.

[0065] In an example, the mixture of agents having silicon dioxide, aluminum oxide, and sodium silicate, beads of the silicon dioxide are significantly larger than the particles of aluminum oxide and sodium silicate.

[0066] The mixture of agents is released from one or more of the cartridge assemblies in response to the thermal event.

[0067] A shape of the container hollow tube **102** can be cylindrical, rectangular, and/or spherical. The cartridge assemblies **100** can, in some examples, after melting, block movement of vent byproducts to help reduce thermal propagation within the battery pack **14**.

[0068] The mixture of ages slows thermal propagation between cells, enhances electrical isolation, lowers vent gas temperatures, and suppresses the thermal event by reducing available oxygen.

[0069] It should be understood that terms such as “about,” “substantially,” and “generally” are not intended to be boundaryless terms, and should be interpreted consistent with the way one skilled in the art would interpret those terms. It should also be understood that directional terms such as “upper,” “top,” “vertical,” “forward,” “rear,” “side,” “above,” “below,” etc., are used herein relative to the normal operational attitude of a vehicle for purposes of explanation only, and should not be deemed limiting.

[0070] Although the different examples have the specific components shown in the illustrations, embodiments of this disclosure are not limited to those particular combinations. It is possible to use some of the components or features from one of the examples in combination with features or components from another one of the examples. In addition, the various figures accompanying this disclosure are not necessarily to scale, and some features may be exaggerated or minimized to show certain details of a particular component or arrangement.

[0071] One of ordinary skill in this art would understand that the above-described embodiments are exemplary and non-limiting. That is, modifications of this disclosure would come within the scope of the claims. Accordingly, the following claims should be studied to determine their true scope and content.

1. A traction battery pack assembly, comprising:
- an enclosure assembly that provides an interior area;
- a cell stack within the interior area, wherein the cell stack includes a plurality of battery cells disposed along a cell stack axis, wherein each of the plurality of battery cells includes at least one terminal tab that projects outward from the cell stack axis;
- a cap; and
- a plurality of fingers projecting from the cap, wherein each of the fingers is spaced-apart from one another to provide at least one slot that receives a portion of the at least one terminal tab, wherein each of the plurality of fingers is provided by a cartridge assembly, wherein each of the cartridge assemblies holds agents, and wherein each of the cartridge assemblies is configured to release the agents in response to a thermal event proximate the respective cartridge assembly.

2. The traction battery pack assembly as recited in claim **1**, wherein the agents are configured, when released from a respective cartridge assembly, to do one or more of (i) arrest oxygen, (ii) electrically isolate, and (iii) reduce temperature.

3. The traction battery pack assembly as recited in claim **2**, wherein each of the cartridge assemblies holds a mixture of silicon dioxide, aluminum oxide, and sodium silicate.

4. The traction battery pack assembly as recited in claim **1**, wherein the cap is an integrally-formed structure without any joints or seams.

5. The traction battery pack assembly as recited in claim **1**, wherein each of the plurality of fingers is formed separately from one another and formed separately from the cap.

6. The traction battery pack assembly as recited in claim **1**, wherein the plurality of fingers includes at least one first finger on a first side of the cell stack, and at least one second finger on an opposite, second side of the cell stack.

7. The traction battery pack assembly as recited in claim **6**, wherein the cap is disposed alongside a third side of the cell stack.

8. The traction battery pack assembly as recited in claim **7**, wherein the third side is transverse to both the first side and the second side.

9. The traction battery pack assembly as recited in claim **1**, wherein each of the cartridge assemblies includes:
- a hollow tube, and
- a container cap adjacent an end of the hollow tube opposite the cap.

10. The traction battery pack assembly as recited in claim **9**, wherein each of the cartridge assemblies is configured such that melting of either the hollow tube or the container cap releases the agents within the respective cartridge assembly.

11. The traction battery pack assembly as recited in claim **10**, wherein:

the cap includes a plurality of male components, and
each of the hollow tubes includes a female component press-fit relative to a corresponding one of the male components.

12. The traction battery pack assembly as recited in claim **1**, wherein the plurality of fingers are a plurality of first fingers disposed along a first side of the cell stack and a plurality of second fingers disposed along a second side of the cell stack.

13. The traction battery pack assembly as recited in claim **1**, wherein the cell stack is a first cell stack, and further comprising at least one second cell stack within the interior area.

14. The traction battery pack assembly as recited in claim **1**, wherein the plurality of battery cells are a plurality of pouch-style battery cells, each pouch-style battery cell having a crimped edge that projects outward from the cell stack axis about a circumferential perimeter of the pouch-style battery cell, wherein the at least one slot receives a portion of at least one of the crimped edge.

15. The traction battery pack assembly of claim **14**, wherein each of the terminal tabs within the plurality of terminal tabs extends through one of the crimped edges.

16. A method, comprising:
placing a cell stack within an interior area of an enclosure of a battery pack;
inserting an assembly including a cap and a plurality of fingers into the interior area, wherein each of the fingers is provided by a cartridge assembly, wherein each of the cartridge assemblies holds agents, and wherein each of the cartridge assemblies is configured to release the agents in response to a thermal event proximate the respective cartridge assembly; and
during the inserting step, receiving at least one tab terminal of the cell stack within a slot between a first finger of the plurality of fingers and a second finger of the plurality of fingers.

17. The method as recited in claim **16**, wherein the agents are configured, when released from a respective cartridge assembly, to do one or more of (i) arrest oxygen, (ii) electrically isolate, and (iii) reduce temperature.

18. The method as recited in claim **17**, wherein each of the cartridge assemblies holds a mixture of silicon dioxide, aluminum oxide, and sodium silicate.

19. The method as recited in claim **16**, wherein the cap is an integrally-formed structure without any joints or seams.

20. The method as recited in claim **16**, wherein the plurality of fingers includes a plurality of fingers on a first side of the cell stack and a plurality of fingers on an opposite, second side of the cell stack.

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